



Geometric and Negative Binomial Distributions · Worksheet 1

Further Statistics 1 Chapter 3: Geometric and Negative Binomial Distributions. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 3. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- $X \sim \text{Geo}(p)$ is the number of trials up to and including the first success; $X \sim \text{NB}(r, p)$ is the number of trials up to and including the r th success. Give probabilities to 4 decimal places.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A The geometric distribution	20	
B Mean, variance and the memoryless property	21	
C The negative binomial distribution	21	
Roar-level challenge	5	
Total	67	

Section A The geometric distribution

20 marks

1. Given that $X \sim \text{Geo}(0.3)$, find $P(X = 4)$. [2]

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2. A fair six-sided dice is rolled repeatedly until a six is obtained. [3]
Find the probability that (a) the first six occurs on the third roll, (b) more than four rolls are needed.

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3. Given that $X \sim \text{Geo}(0.2)$, find (a) $P(X \leq 5)$, (b) $P(3 \leq X \leq 6)$. [3]

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4. The random variable $X \sim \text{Geo}(p)$, where $p < 0.5$. Given that $P(X = 2) = 0.21$, [4]

(a) find the value of p ,

(b) find $P(X > 3)$.

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5. An archer hits the centre of a target with probability 0.35 on each shot, independently of all other shots. She shoots until she first hits the centre. The random variable X is the number of shots she takes.

(a) Write down the distribution of X . [1]

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(b) Find $P(X = 3)$. [2]

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(c) Find the probability that she needs more than 5 shots. [2]

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(d) Find the smallest value of n such that $P(X \leq n) > 0.95$. [3]

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(Total for Question 5 is 8 marks)



Section B Mean, variance and the memoryless property

21 marks

6. Given that $X \sim \text{Geo}(0.25)$, find $E(X)$ and $\text{Var}(X)$. [2]

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7. The random variable X has a geometric distribution with mean 5. [3]
Find the variance and the standard deviation of X .

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8. Given that $X \sim \text{Geo}(0.4)$, find $P(X > 7 \mid X > 4)$, showing your working. [3]

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9. The random variable $X \sim \text{Geo}(p)$ has variance 6. [4]

(a) Show that $6p^2 + p - 1 = 0$ and hence find p .

(b) Find $P(X = E(X))$.

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10. Each packet of a brand of cereal contains a toy. The probability that a packet contains a rare toy is 0.08, independently of all other packets. Sam buys packets one at a time until she gets a rare toy.

(a) Find the mean and the standard deviation of the number of packets Sam buys. [2]

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(b) Find the probability that Sam buys more than 10 packets. [2]

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(c) Sam has bought 6 packets and none contained a rare toy. Find the probability that she needs to buy more than 16 packets in total, and name the property of the geometric distribution that you have used. [3]

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(d) Each packet costs £3. Find the mean and the standard deviation of the total amount Sam spends. [2]

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(Total for Question 10 is 9 marks)



Section C The negative binomial distribution

21 marks

11. Given that $X \sim \text{NB}(3, 0.4)$, find $P(X = 5)$. [2]

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12. Given that $X \sim \text{NB}(4, 0.5)$, find $E(X)$, $\text{Var}(X)$ and $P(X = 6)$. [3]

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13. A fair six-sided dice is rolled repeatedly. Find the probability that the second six occurs on the fifth roll. [3]

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14. A basketball player scores each free throw with probability 0.7, independently. Find the probability [4]
that
(a) her third score comes on her fifth attempt,
(b) she has scored three times within her first four attempts.

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15. A salesperson makes telephone calls. Each call, independently, results in a sale with probability 0.2. On one day she keeps calling until she has made 3 sales. The random variable X is the number of calls she makes.

(a) Find $P(X = 10)$. [2]

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(b) Find $E(X)$ and $\text{Var}(X)$. [2]

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(c) Find the probability that she needs more than 5 calls. [3]

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(d) Given that exactly one of her first 5 calls results in a sale, find the probability that she makes exactly 10 calls. [2]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Two players, A and B , take turns to roll a fair six-sided dice, with A rolling first. The winner is the first player to roll a six. [5]

- (a) Show that the probability that A wins is $\frac{6}{11}$.
(b) Find the probability that A wins on or before his third roll.

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END OF WORKSHEET

Total: 67 marks